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Operating System

PRODUCER CONSUMER USING SEMAPHORES

Aim: To write a program to implement solution to producer consumer problem using semaphores.

Algorithm:

1. Initialize semaphore empty, full and mutex.
2. Create two threads- producer thread and consumer thread.
3. Wait for target thread termination.
4. Call sem_wait on empty semaphore followed by mutex semaphore before entry into critical section.
5. Produce/Consume the item in critical section.
6. Call sem_post on mutex semaphore followed by full semaphore
7. before exiting critical section.
8. Allow the other thread to enter its critical section.
9. Terminate after looping ten times in producer and consumer Threads each.

Program Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <semaphore.h>

#define SIZE 3

int buffer[SIZE];
int in = 0, out = 0, count = 0;

sem_t empty, full, mutex;

void producer() {
    if (count == SIZE) {
        printf("Buffer is full!!\n");
        return;
    }

    sem_wait(&empty);
    sem_wait(&mutex);

    buffer[in] = count + 1;
```

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```
printf("Producer produces the item %d\n", buffer[in]);

in = (in + 1) % SIZE;
count++;

sem_post(&mutex);
sem_post(&full);
}

void consumer() {
    if (count == 0) {
        printf("Buffer is empty!!\n");
        return;
    }

    sem_wait(&full);
    sem_wait(&mutex);

    int item = buffer[out];
    printf("Consumer consumes item %d\n", item);

    out = (out + 1) % SIZE;
    count--;

    sem_post(&mutex);
    sem_post(&empty);
}

int main() {
    int choice;

    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);
    sem_init(&mutex, 0, 1);

    while (1) {
        printf("\n1. Producer\n2. Consumer\n3. Exit\n");
        printf("Enter your choice:");
        scanf("%d", &choice);

        switch (choice) {
```

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```
        case 1:
            producer();
            break;
        case 2:
            consumer();
            break;
        case 3:
            exit(0);
        default:
            printf("Invalid choice!\n");
    }
}

return 0;
}
```

Sample Output:

```
1. Producer
2.Consumer
3.Exit
Enter your choice:1
Producer produces the item 1
Enter your choice:2
Consumer consumes item
1 Enter your choice:2
Buffer is empty!!
Enter your choice:1
Producer produces the item 1
Enter your choice:1
Producer produces the item 2
Enter your choice:1
Producer produces the item 3
Enter your choice:1
Buffer is full!!
Enter your choice:3
```

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Result:

```
main.c  [Icons]  Share  Run  Output

1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <pthread.h>
4  #include <semaphore.h>
5
6  #define BUFFER_SIZE 3
7
8  sem_t empty, full, mutex;
9  int buffer[BUFFER_SIZE];
10 int in = 0, out = 0;
11 int item = 0;
12
13 // Producer function
14 void* producer(void* arg) {
15     if(sem_trywait(&empty) != 0) {
16         printf("Buffer is full!!\n");
17         return NULL;
18     }
19
20     sem_wait(&mutex);
21
22     item++;
23     buffer[in] = item;
24     printf("Producer produces the item %d\n", item);
25     in = (in + 1) % BUFFER_SIZE;
26
27     sem_post(&mutex);
28     sem_post(&full);
29
30     return NULL;
31 }
32
33 // Consumer function
34 void* consumer(void* arg) {
```

1. Producer
2. Consumer
3. Exit
Enter your choice:1
Producer produces the item 1

1. Producer
2. Consumer
3. Exit
Enter your choice:2
Consumer consumes item 1

1. Producer
2. Consumer
3. Exit
Enter your choice:2
Buffer is empty!!

1. Producer
2. Consumer
3. Exit
Enter your choice:|

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```
35- if(sem_trywait(&full) != 0) {
36-     printf("Buffer is empty!!\n");
37-     return NULL;
38- }
39-
40- sem_wait(&mutex);
41-
42- int consumed = buffer[out];
43- printf("Consumer consumes item %d\n", consumed);
44- out = (out + 1) % BUFFER_SIZE;
45-
46- sem_post(&mutex);
47- sem_post(&empty);
48-
49- return NULL;
50- }
51-
52- int main() {
53-     int choice;
54-     pthread_t p, c;
55-
56-     // Initialize semaphores
57-     sem_init(&empty, 0, BUFFER_SIZE);
58-     sem_init(&full, 0, 0);
59-     sem_init(&mutex, 0, 1);
60-
61-     while(1) {
62-         printf("\n1. Producer\n2. Consumer\n3. Exit\n");
63-         printf("Enter your choice:");
64-         scanf("%d", &choice);
65-
66-         switch(choice) {
67-             case 1:
68-                 pthread_create(&p, NULL, producer, NULL);
```

1. Producer
2. Consumer
3. Exit
Enter your choice:1
Producer produces the item 1

1. Producer
2. Consumer
3. Exit
Enter your choice:2
Consumer consumes item 1

1. Producer
2. Consumer
3. Exit
Enter your choice:2
Buffer is empty!!

1. Producer
2. Consumer
3. Exit
Enter your choice:

```
53- int choice;
54- pthread_t p, c;
55-
56- // Initialize semaphores
57- sem_init(&empty, 0, BUFFER_SIZE);
58- sem_init(&full, 0, 0);
59- sem_init(&mutex, 0, 1);
60-
61- while(1) {
62-     printf("\n1. Producer\n2. Consumer\n3. Exit\n");
63-     printf("Enter your choice:");
64-     scanf("%d", &choice);
65-
66-     switch(choice) {
67-         case 1:
68-             pthread_create(&p, NULL, producer, NULL);
69-             pthread_join(p, NULL);
70-             break;
71-
72-         case 2:
73-             pthread_create(&c, NULL, consumer, NULL);
74-             pthread_join(c, NULL);
75-             break;
76-
77-         case 3:
78-             exit(0);
79-
80-         default:
81-             printf("Invalid choice\n");
82-     }
83- }
84-
85- return 0;
86- }
```

1. Producer
2. Consumer
3. Exit
Enter your choice:1
Producer produces the item 1

1. Producer
2. Consumer
3. Exit
Enter your choice:2
Consumer consumes item 1

1. Producer
2. Consumer
3. Exit
Enter your choice:2
Buffer is empty!!

1. Producer
2. Consumer
3. Exit
Enter your choice: